

Finite Element Analysis of a Cantilever Beam Using 2D Four-Node Rectangular Elements

1. Introduction

The objective of this project is to develop a two-dimensional finite element method (FEM) code for the static and modal analysis of a cantilever beam using four-node rectangular elements. The beam is modeled as a rectangular plane stress domain subjected to distributed transverse loading. The study includes mesh convergence analysis, comparison with Euler–Bernoulli and Timoshenko beam theories, investigation of aspect ratio effects, stress and strain visualization, and modal analysis.

The project is divided into two parts:

1. Static analysis of a cantilever beam subjected to distributed transverse traction.
 2. Modal analysis for determination of natural frequencies and mode shapes.
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PART I: STATIC ANALYSIS

2. Problem Description

A rectangular cantilever beam of length (L_x) and height (L_y) is considered. The beam thickness (t) is small compared to other dimensions and the structure is modeled using plane stress assumptions.

One end of the beam is fixed while a distributed transverse traction is applied along the upper boundary.

Material Properties

- Young's modulus: $E = 210 \cdot 10^9$; Pa
- Poisson's ratio $\nu = 0.3$
- Thickness: t

3. Finite Element Formulation

3.1 Element Type

The domain is discretized using four-node bilinear rectangular elements.

Each node possesses two degrees of freedom u_x, u_y

Thus each element has: 8 DOF

3.2 Plane Stress Constitutive Matrix

The constitutive matrix for plane stress is:

$$D = \frac{E}{1 - \nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix}$$

3.3 Element Stiffness Matrix

The element stiffness matrix is computed using:

$$K_e = \int_A B^T D B t dA$$

where:

- B = strain-displacement matrix
- D = constitutive matrix
- t = thickness

Numerical integration is performed using Gauss quadrature.

4. Mesh Generation

The finite element code was developed with the following capabilities:

- Input of beam dimensions (L_x) and (L_y)
- Variable number of elements in x and y directions
- Automatic generation of node coordinates
- Automatic element connectivity generation

Mesh Generation Function

The function `generate_mesh( $L_x$ ,  $L_y$ ,  $n_x$ ,  $n_y$ )` was used to generate the finite element mesh.

The mesh spacing is: $dx = L_x/n_x$ $dy = L_y/n_y$

The beam aspect ratio is defined as: $AR = L_x/L_y$

while element aspect ratio is: $AR = dx/dy$

5. Boundary Conditions and Loading

5.1 Boundary Conditions

The left edge of the beam was fully constrained.

Thus for all nodes satisfying: $x = 0$ both displacement components were fixed: $u_x = 0$ $u_y = 0$

5.2 Distributed Traction Loading

A distributed transverse traction was applied on the upper boundary.

The equivalent nodal force vector was assembled into the global force vector.

6. FEM Solution Procedure

The global system of equations is: $KU = F$

where:

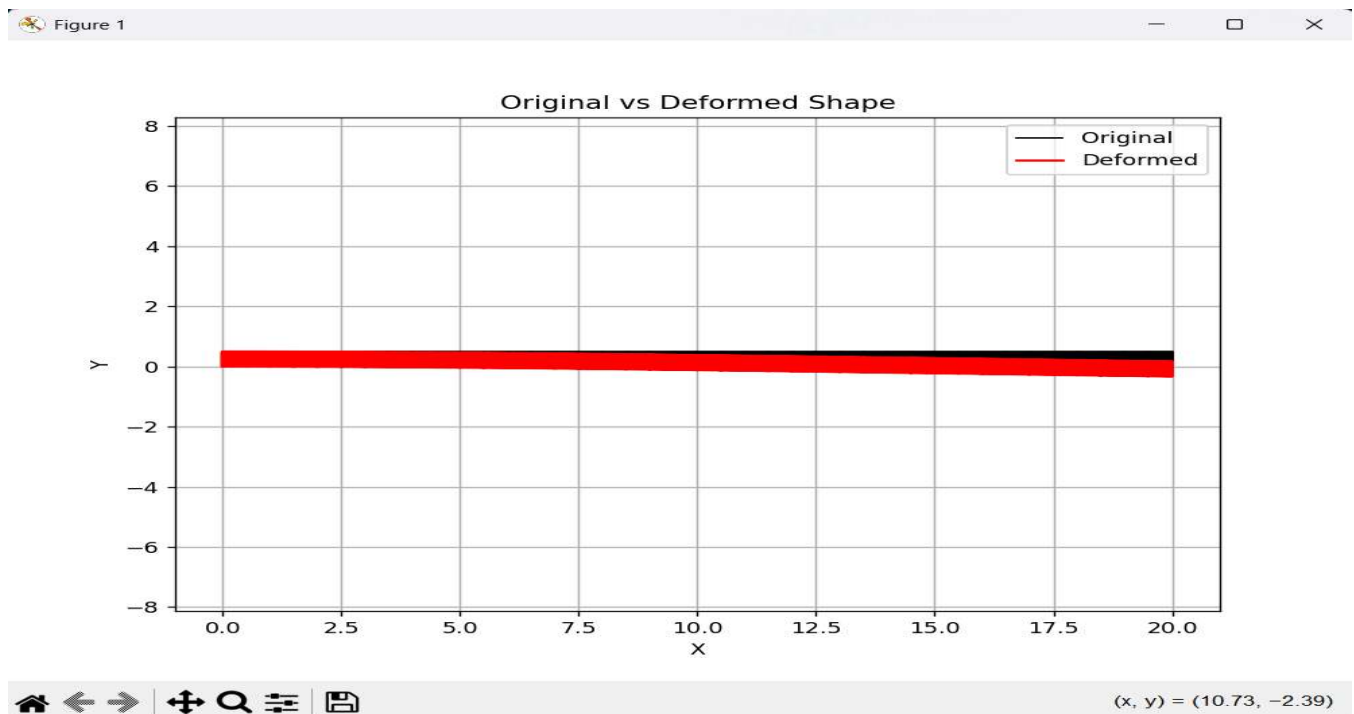
- K = global stiffness matrix
- U = global displacement vector
- F = global force vector

After applying boundary conditions: $K_r U_r = F_r$

- K_r = global stiffness matrix
- U_r = global displacement vector
- F_r = global force vector

The reduced system was solved using NumPy linear solvers.

```
Enter length in x-direction: 20
Enter length in y-direction: .5
Enter number of elements in x-direction: 40
Enter number of elements in y-direction: 40
```



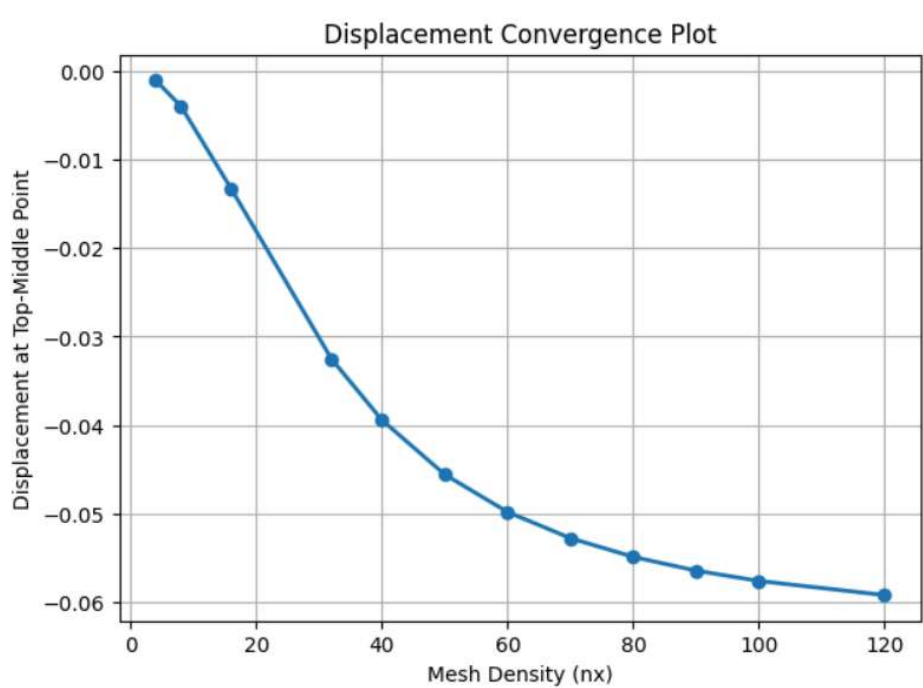
7. Mesh Convergence Analysis

7.1 Objective

Mesh convergence analysis was performed for beam aspect ratio: $AR = 50$
The following quantities were monitored:

1. Total strain energy
2. Displacement at point A

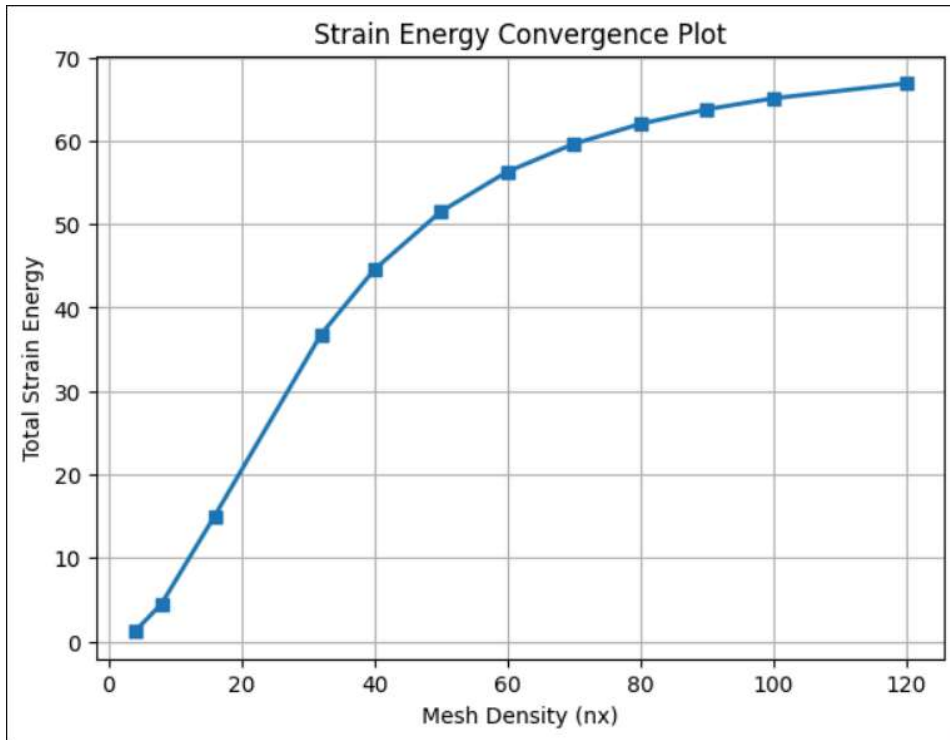
The mesh was progressively refined using increasing values of n_x .



7.2 Strain Energy

$$U_{strain} = \frac{1}{2} U^T K U$$

The total strain energy was computed using: $U_{strain} =$



Convergence was achieved when further mesh refinement produced negligible change in Displacement at A ,

7.3 Displacement Monitoring

Displacement was evaluated at point A located at the midpoint of the free end: $(x,y) = (Lx, Ly/2)$

This location corresponds to the beam neutral axis and provides proper comparison with analytical beam theories.

8. Validation with Beam Theory

The FEM solution was validated using:

1. Euler–Bernoulli beam theory
2. Timoshenko beam theory

8.1 Euler–Bernoulli Beam Theory

Euler–Bernoulli theory assumes:

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$$\gamma_{xy} = 0$$

Thus shear deformation is neglected.

The tip deflection for uniformly distributed load is:

$$\delta_{EB} = \frac{wL^4}{8EI}$$

where:

$$I = \frac{tL_y^3}{12}$$

8.2 Timoshenko Beam Theory

Timoshenko beam theory includes shear deformation effects.

The tip deflection is:

Timoshenko beam theory includes shear deformation effects.

The tip deflection is:

$$\delta_T = \frac{wL^4}{8EI} + \frac{wL^2}{2kGA}$$

where:

$$G = \frac{E}{2(1 + \nu)}$$

and:

$$k = \frac{5}{6}$$

Euler–Bernoulli strain energy:

$$U_{EB} = \frac{w^2 L_x^5}{40EI}$$

Timoshenko strain energy:

$$U_T = \frac{w^2 L_x^5}{40EI} + \frac{w^2 L_x^3}{6kGA}$$

where:

- w = distributed load
- L_x = beam length
- E = Young's modulus
- I = second moment of area
- k = shear correction factor
- G = shear modulus
- A = cross-sectional area

Code results for Nx=120 and Ny =120

```
FEM = -0.16715638520348391
Euler-Bernoulli = -0.17857142857142852
Timoshenko = -0.17864571428571424
EB Error % = 6.392424286048981
Timoshenko Error % = 6.431348844929493
```

```
Strain energy Fem 66.8674468558886
Euler-Bernoulli: 71.42857142857142
Timoshenko: 71.47809523809522
EB Error %: 6.3855744017559415
Timoshenko Error %: 6.450435433188925
```

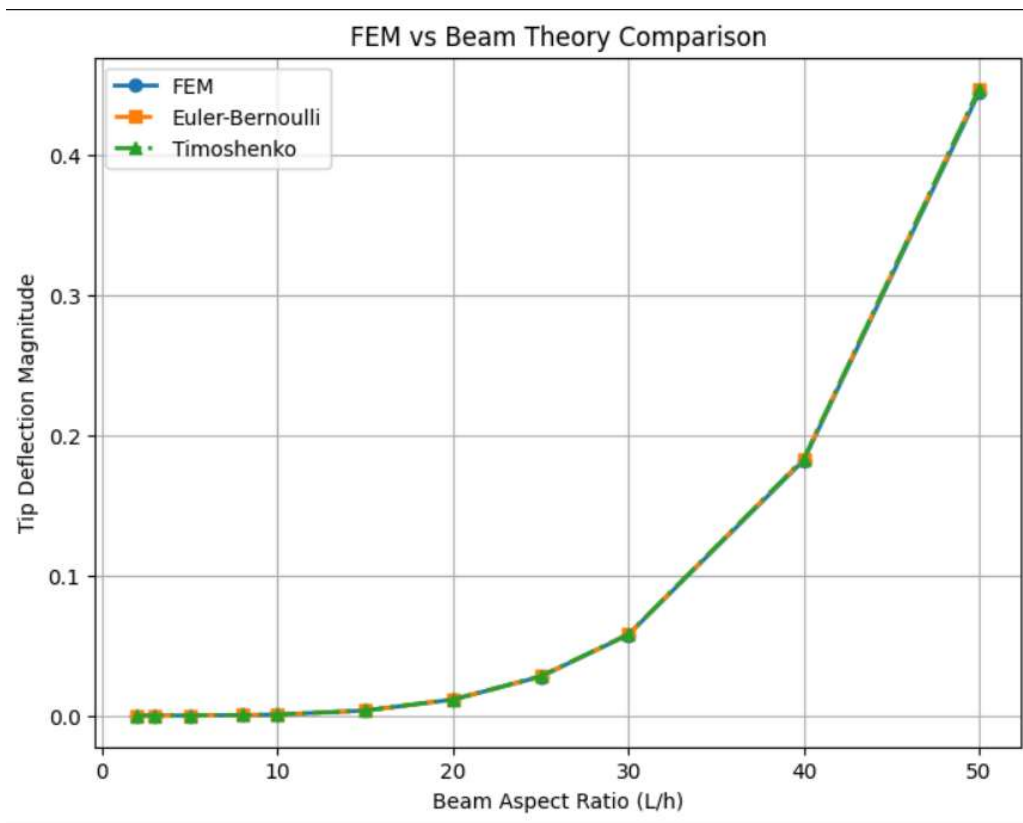
9. Aspect Ratio Investigation

9.1 Objective

A case study was performed for beam aspect ratios:

[2, 3, 5, 8, 10, 15, 20, 25, 30, 40, 50]

The converged mesh density was maintained while varying beam geometry.



9.2 Observations

Small Aspect Ratio (Thick Beam)

For small aspect ratios: L less than 5
significant shear deformation was observed.

Observations:

- Euler-Bernoulli theory underestimated deflection.

- Timoshenko theory showed much better agreement with FEM.
 - Shear effects were important.
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Large Aspect Ratio (Slender Beam)

for large aspect ratios: L greater than 20 shear deformation became negligible.

Observations:

- FEM, Euler–Bernoulli, and Timoshenko solutions converged.
 - Beam behavior became bending dominated.
 - Euler–Bernoulli assumptions became valid.
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10. Shear Stress Distribution

Shear stress through thickness was evaluated at: $x = L/2$

For: Beam AR = 50

The FEM solution showed approximately parabolic variation.

For a rectangular beam:

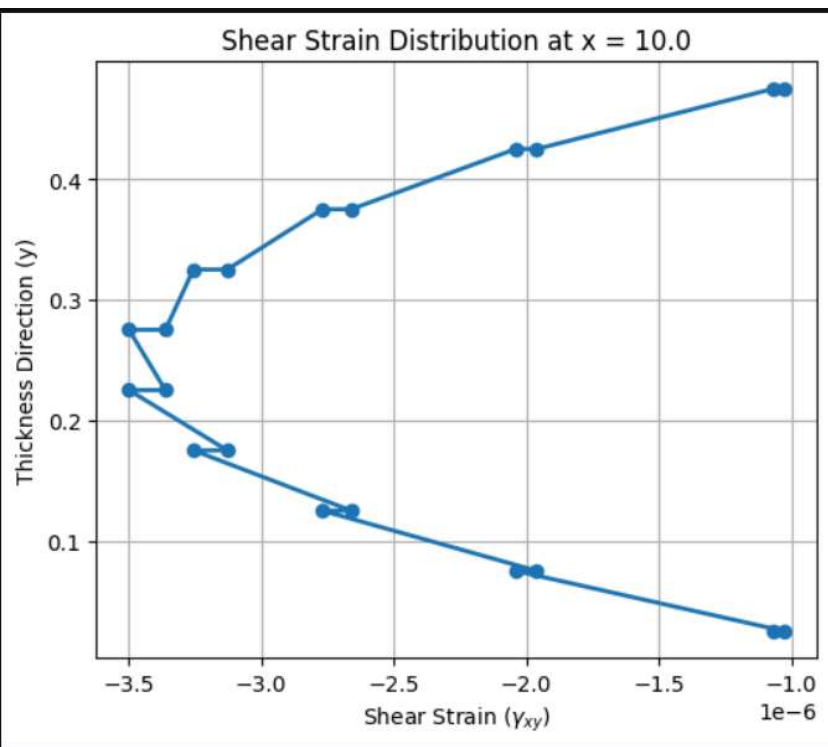
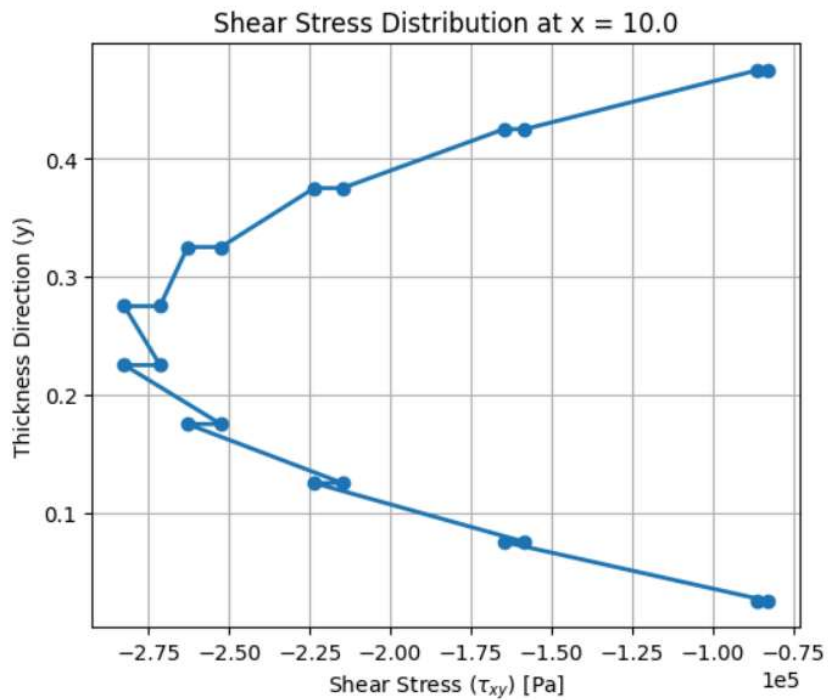
$$\tau_{xy}(y) = \frac{3V}{2A} \left(1 - \frac{4y^2}{h^2} \right)$$

The following observations were made:

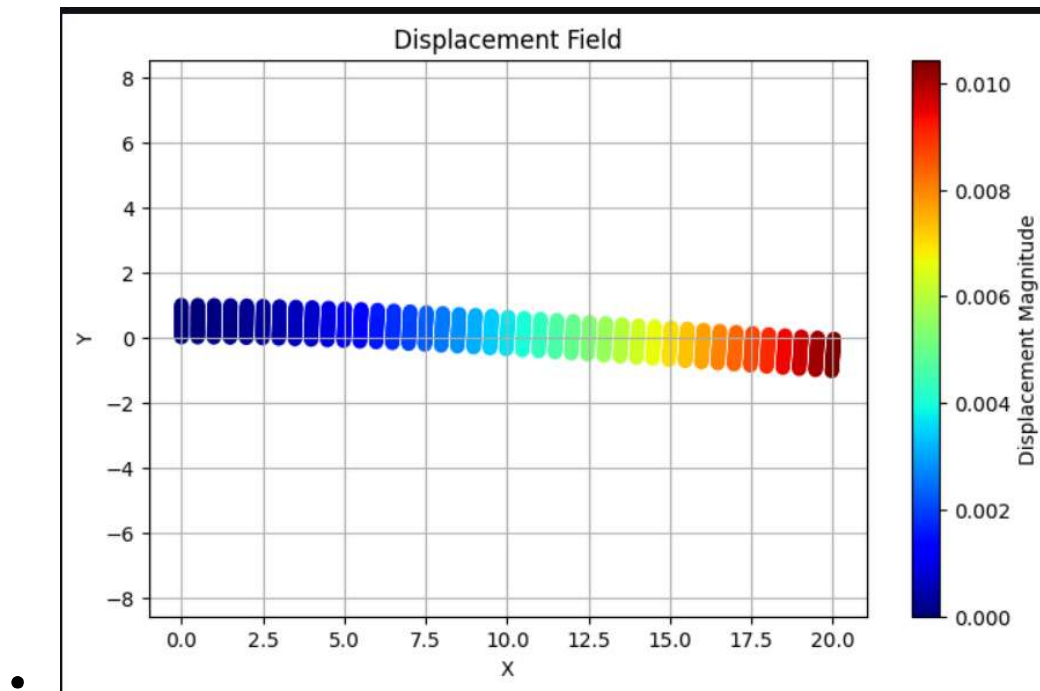
- Shear stress was maximum near the neutral axis.
 - Shear stress approached zero near top and bottom surfaces.
 - FEM results agreed qualitatively with elasticity theory.
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11. Stress and Strain Fields

The following contour plots were generated:



- Displacement field



- Shear stress field
- Shear strain field

The deformed shape of the cantilever beam clearly showed bending behavior.

Stress concentration was observed near the fixed boundary.

PART II: MODAL ANALYSIS

12. Objective

Modal analysis was performed for: beam AR = 50

The objective was to compute:

1. First five natural frequencies
2. Corresponding mode shapes

13. Finite Element Formulation for Modal Analysis

The governing eigenvalue problem is:

$$(K - \omega^2 M)\phi = 0$$

where:

- (K) = global stiffness matrix
- (M) = global mass matrix
- (ω) = natural frequency
- (ϕ) = mode shape

The mass matrix was assembled using consistent mass formulation.

14. Mesh Convergence for Modal Analysis

Mesh convergence analysis was performed for the 5th natural frequency.

The mesh density was progressively refined until frequency convergence was achieved.

It was observed that:

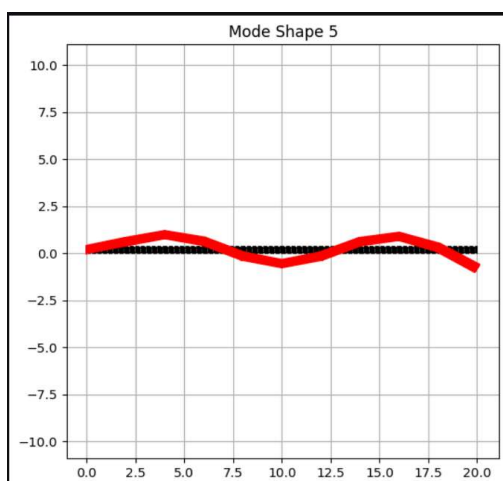
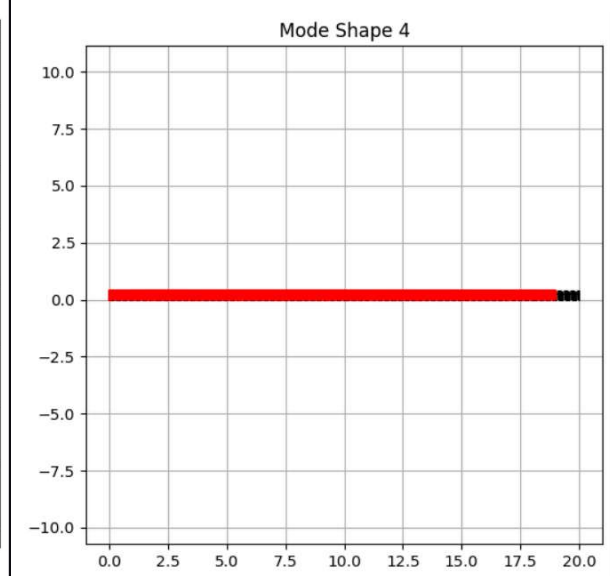
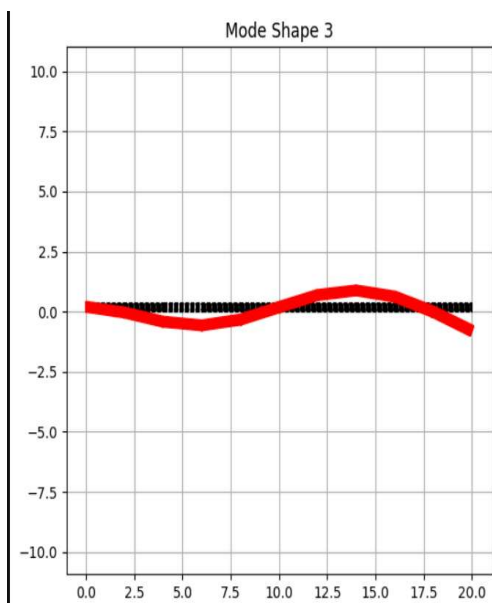
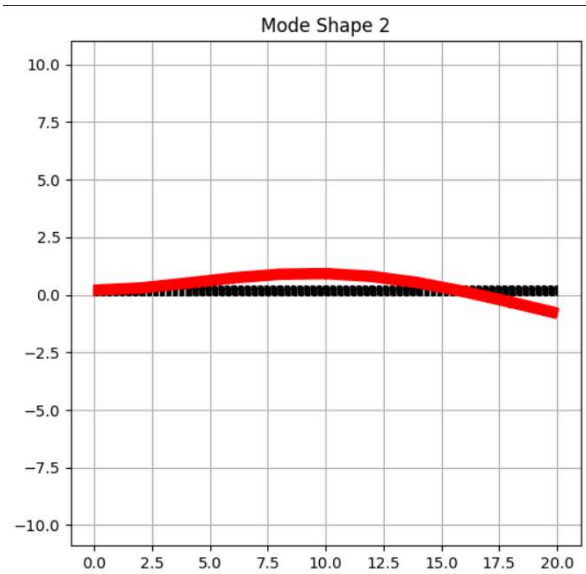
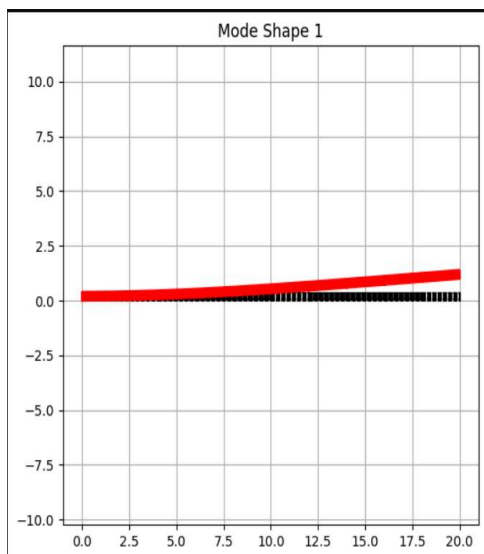
- Coarse meshes produced noticeable errors.
 - Higher modes required finer meshes for convergence.
 - Frequency values stabilized with mesh refinement.
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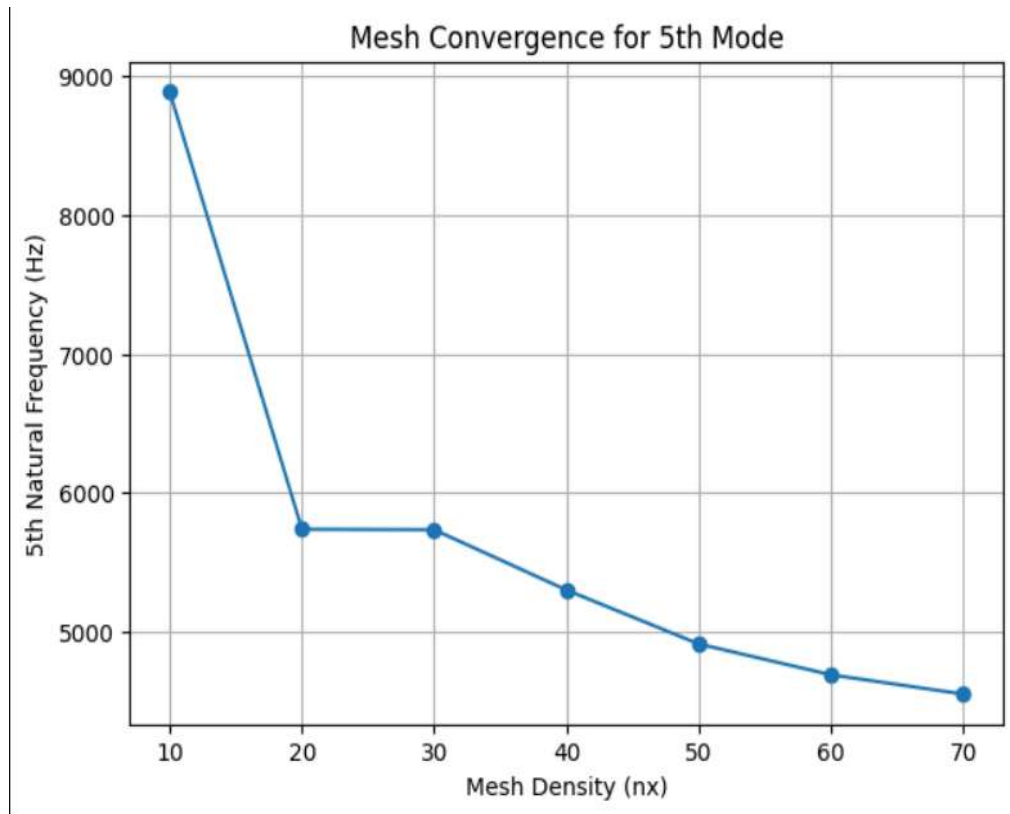
15. Mode Shapes

The first five mode shapes were plotted.

Observations:

- First mode corresponded to fundamental bending.
- Higher modes showed multiple curvature changes.
- Mode complexity increased with mode number.





16. Conclusions

The following conclusions were obtained from this project:

1. The developed FEM code successfully modeled static and modal behavior of cantilever beams using 4-node rectangular elements.
 2. Mesh refinement significantly improved solution accuracy.
 3. For slender beams, Euler–Bernoulli and Timoshenko beam theories closely matched FEM results.
 4. For thick beams, Timoshenko beam theory provided much better agreement due to inclusion of shear deformation.
 5. Shear stress distribution through thickness followed the expected parabolic trend.
 6. Modal analysis successfully predicted natural frequencies and mode shapes.
 7. Higher vibration modes required finer meshes for convergence.
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17. Future Scope

Possible future extensions include:

- Use of higher-order elements
- Plane strain formulation
- Dynamic transient analysis
- Geometric nonlinearity
- Material nonlinearity
- Adaptive mesh refinement